MULTIMODE COUNTER

FUNCTIONALITY

Upon power-up, the system initializes to zero. Interact via Nexys buttons to reset, pause, or load 8-bit data. Real-time switching between Decimal, Hexadecimal, and Custom Scroll patterns occurs without losing the count. 16 LEDs show raw binary state, while the 7-segment display provides readable feedback.

SYSTEM FEATURES

- 100MHz synchronous design
- Debounced button processing
- Adjustable step sizes (1-15)
- Four auto-counting speeds
- Real-time data loading
- 8-digit multiplexed formatting

OUTPUT MODES

DECIMAL

```
00000000 to 00065535
```

HEXADECIMAL

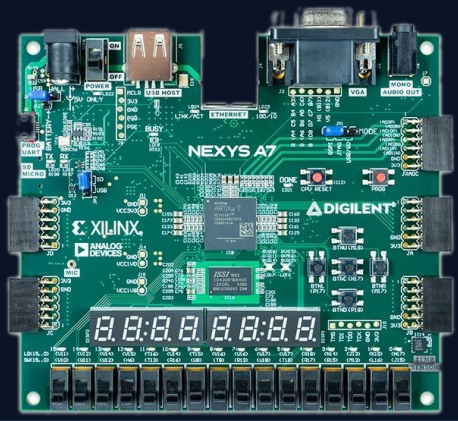
```
00000000 to 0000FFFF
```

SCROLL

```
Custom "DE1" Pattern
```

I/O MAPPING

Input	Function
BTNC	Reset Counter (to 0)
BTNR	Cycle Display Mode
BTNL	Run/Pause Auto Count
BTNU	Single Manual Step
SW[0]	Direction (0=Up, 1=Dn)
SW[7:1]	Tick Freq Select
SW[15:8]	Set Step Size (1-15)
LD[15:0]	Loadable Data Vector
DISP[7:0]	7-Segment Output



HARDWARE ARCHITECTURE FLOW

Switches & Buttons → Debounce & Clock Gen → 16-Bit Counter & Control → Display / LEDs

VALUE AND IMPACT

Demonstrates synchronous hardware design reliability. Displaying values up to **65,535** across multiple formats, it highlights FPGA efficiency in high-speed timing versus traditional microcontrollers.